NEXUS: a dependency-free drug-discovery pipeline, and an account of what it does and does not do

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Abstract

NEXUS is a drug-discovery pipeline written against NumPy alone: 7,194 lines implementing SMILES parsing and ring perception, molecular fingerprints and descriptors, reverse-mode automatic differentiation, a directed message-passing network, Gaussian processes and active learning, split-conformal uncertainty, persistent homology, discrete Ricci curvature, non-equilibrium free-energy estimators, a discrete graph diffusion model, and a Gene Ontology layer. It covers hit identification, lead optimisation, binding-affinity prediction and de novo generation, and ships as a panel for the Mantis map interface.

The point of the exercise is not that any one component is novel. It is that every component is checked against something external — published descriptor tables, analytically solvable free energies, finite-difference gradients, known topology, hand-computed hypergeometric tails — and that the results are reported whichever way they come out. A D-MPNN reproduces published MoleculeNet numbers, loses to a plain fingerprint baseline on the smaller datasets, and wins on the largest by beating both the baselines and the published figure — the regime crossover made visible rather than asserted. Conformal intervals attain their guaranteed coverage, and the adaptive variant that should improve on them helps on one dataset of three and the textbook construction is worse on all of them. The generator first scored *below* its own untrained control; that was two bugs, and fixing them took it to 64.7% validity against a 2.0% control, with every remaining failure a single nameable gap. Each of these is reported here as measured, because a pipeline that only produces good numbers is not a pipeline anyone can use.

1 Introduction

Computational drug discovery runs on a deep software stack. RDKit parses the structures, scikit-learn or PyTorch fits the models, SciPy supplies the statistics, and a dozen smaller packages fill the gaps. The stack is excellent and nobody should rewrite it for production work. But it also means that a practitioner can run a screening campaign end to end without once being forced to confront what a Murcko scaffold is, where an aromaticity model draws its line, or what exactly a confidence interval from a Gaussian process guarantees. The abstractions are good enough to be used without being understood, and that is usually a virtue.

NEXUS is the opposite exercise. It implements the pipeline against NumPy alone — 7,194 lines of library code, 2,218 lines of tests, 298 tests — specifically so that every modelling decision has to be made explicitly and can therefore be checked. SMILES parsing, ring perception and Hückel aromaticity; Bemis–Murcko scaffolds [3]; Morgan fingerprints [16] and Ertl TPSA [9]; a tape-based reverse-mode autodiff engine and a directed message-passing network [20]; Gaussian-process active learning; split-conformal prediction [18, 11]; Vietoris–Rips persistent homology [8]; Ollivier–Ricci curvature [12]; Jarzynski, Crooks and BAR free-energy estimators [10, 6, 5]; a discrete denoising diffusion model for molecular graphs [2, 17]; and a Gene Ontology layer with information-theoretic semantic similarity [1, 15].

What this paper claims. Not that these implementations are faster or more accurate than the established ones — they are neither. The claim is narrower and, we think, more useful: that writing them out and validating each against an external reference exposes failures that a stack of trusted packages would have hidden. Three such failures are documented here. A scaffold splitter silently handed the validation set zero molecules on one dataset. A graph generator scored below its own untrained baseline because 83.7% of atom pairs are "no bond" and the denoiser had learned to say so. A conformal interval attained exactly its guaranteed coverage while being the same width for every molecule, which is correct and nearly useless. None of these would have produced an error message.

What this paper does not claim. There is no state-of-the-art result here. Where a baseline beats the neural model, the table says so. Where an improvement fails to improve, the negative result is reported with the same prominence as the positive ones (Section 4). Hyperparameters selected on a reported metric are flagged as such.

2 Method

2.1 Chemistry core

Structures are parsed from an OpenSMILES subset into an explicit atom/bond graph. Rings are perceived by Horton's smallest-set-of-smallest-rings algorithm with independence checked over GF(2); aromaticity follows a Hückel $4n + 2$ rule restricted by a conservative Kekulé pass, so a ring carrying an exocyclic double bond — the carbonyl ring of

caffeine is the canonical case — is not aromatised. Implicit hydrogens follow a standard valence model, and a molecule that cannot satisfy it is reported as invalid rather than silently repaired.

Descriptors are computed from this graph: molecular weight, heavy-atom and ring counts, rotatable bonds, hydrogen-bond donors and acceptors, fraction of sp^3 carbon, Ertl topological polar surface area [9], and a Crippen-style logP. Morgan/ECFP fingerprints [16] are hashed from iteratively refined atom environments. Scaffolds follow Bemis–Murcko [3].

2.2 Learning

A tape-based reverse-mode autodiff engine over NumPy arrays supplies gradients; every operation is checked against central finite differences. On top of it sits a directed message-passing network [20], where messages live on directed bonds and the reverse bond is subtracted at each aggregation to stop a message returning along the edge it arrived on.

Uncertainty is split-conformal [18, 11]. The calibration quantile is the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of the calibration scores — the finite-sample correction, not the empirical quantile at $1-\alpha$, which under-covers at small n. With an absolute-residual score every interval has identical width; scaling the score by a per-molecule difficulty estimate preserves the marginal guarantee while letting width vary, and Section 3.5 measures whether that is worth doing.

2.3 Geometry, physics and generation

Conformers are embedded by distance geometry with triangle smoothing, then relaxed under an explicit out-of-plane term: distance restraints alone constrain planarity only to second order, and benzene puckers without it. Persistent homology is computed on the Vietoris–Rips filtration with a $\mathbb{Z}/2$ boundary-matrix reduction. Ollivier–Ricci curvature [12] uses an exact transportation simplex rather than an entropic approximation.

Both shape descriptions are exposed through the Mantis panel rather than left in the library, which is what Section 3.4 measures.

Free energies come from Jarzynski's equality, the Crooks fluctuation theorem and BAR [10, 6, 5], with BAR solved in a form whose residual is monotone in ΔF, so bisection is guaranteed to find the unique root.

Generation is discrete denoising diffusion over atom and bond categories [2, 17], with marginal transition matrices and a cosine schedule. Reverse diffusion starts from the marginal noise prior; bond choices that would exceed an atom's valence are masked at every reverse step.

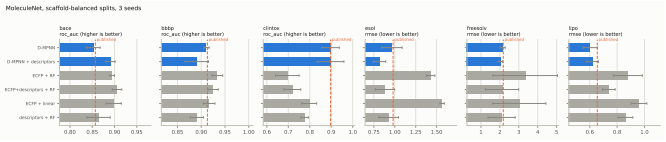


Figure 1: MoleculeNet [19] under scaffold-balanced splits [20], three seeds, with seed-to-seed spread. The neural model is competitive but not dominant.

3 Results

Every number in this section is generated from a file in `results/` by `benchmarks/paper_numbers.py`; none is typed into the text. Figures are rendered from the same files.

3.1 Component validation

Each component is checked against an external reference before it is used for anything. Descriptors are compared against reference columns distributed with the benchmark datasets, over 16,231 molecules, with the best agreement reaching 100.0% exact. Free-energy estimators are run on an analytically solvable switching problem where ΔF is known in closed form. Persistent homology is run on point clouds of known topology — a circle must yield exactly one long-lived H_1 feature, acyclic octane none. Every autodiff operation is checked against central finite differences. The Gene Ontology layer loads 48,340 terms and 904,644 annotations, recovers known pathway enrichments as positive controls, and on 40 random gene sets of 60 genes returns a mean of 0.28 significant terms, with any hit at all in 7.5% of trials — the false-positive behaviour the Benjamini–Hochberg procedure [4] promises. The enrichment the Mantis panel runs is a separate, pruned index and is validated separately (Section 3.6).

3.2 Property prediction

The D-MPNN reproduces published numbers under the same protocol: BACE 0.853 against a published 0.857, BBBP 0.911 against 0.913, ClinTox 0.896 against 0.897 [20]. That is the check that the implementation is right.

It is not the headline. On BACE a plain ECFP-plus-descriptor random forest reaches 0.906, and on BBBP an ECFP random forest reaches 0.934 — both ahead of the message-passing network, on datasets of one to two thousand molecules.

Where the neural model wins is where the regime argument says it should. On Lipophilicity, the largest set here at 4 200 molecules, the D-MPNN reaches 0.603 RMSE: ahead of every fingerprint baseline (0.739 for the best of them) and ahead of the published 0.655 for the same architecture and protocol [20]. It also wins on ClinTox (0.896 against 0.695). A learned representation needs data to pay for itself, and across these six datasets the crossover is visible rather than asserted.

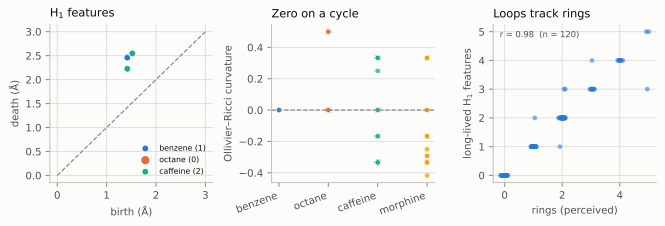


Figure 2: Left: H_1 features of three conformers, all far above the diagonal. Centre: Ollivier–Ricci curvature per bond; benzene is a single point on zero. Right: long-lived H_1 features against perceived ring count.

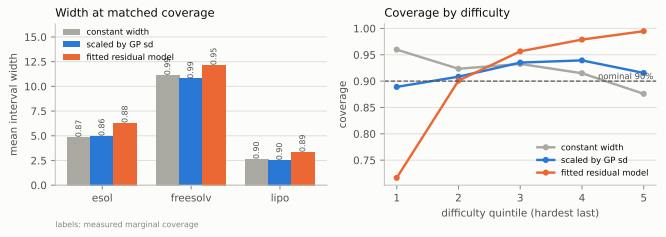


Figure 3: Left: mean interval width, labelled with the marginal coverage each score achieved. Right: coverage within difficulty quintiles. Flat at nominal is ideal.

Both halves are reported because quoting only the dataset where the fancier model wins is how benchmark tables become untrustworthy.

3.3 Binding affinity

On the Davis kinase panel [7], 30,056 measured pairs over 68 drugs and 442 targets, a random split gives concordance index 0.853 and Pearson 0.750, against roughly 0.878 for DeepDTA on the same dataset [13].

The number that matters for prospective use is the other one. Held out by *drug* — the realistic case, where the compound has never been assayed against anything — concordance falls to 0.679 and Pearson to 0.388. Held out by target it falls less far, to 0.820. A random split on an interaction matrix measures interpolation between known drug–target pairs, not the generalisation a screening campaign actually needs, and the gap between 0.853 and 0.679 is the size of that distinction.

3.4 Shape: topology and curvature

The pipeline computes two shape descriptions that no descriptor in Section 3.2 captures, and both are checked against something that is true independently of the code.

Persistent homology is computed on the 3D conformer and never sees the bond graph. On 5 probe structures it recovers the expected number of long-lived H_1 features in 5 of 5 cases — benzene one, acyclic octane none, caffeine and naphthalene two. The interesting test is the one that is not hand-picked: across 198 molecules from ESOL, the count of features surviving more than 0.35 Å correlates with the ring count perceived from connectivity at $r = 0.979$, and agrees *exactly* on 94.9% of them (mean absolute error 0.056 rings). Geometry alone recovers a combinatorial property of the molecular graph.

Ollivier–Ricci curvature is computed on the bond graph with an exact transportation simplex. For a cycle graph C_n with $n \geq 6$ the curvature is exactly zero, which is a theorem rather than a fit; on 5 ring sizes the implementation returns 0.00×10^0 on every bond, in 5 of 5 cases. That exactness is what makes the quantity readable in the panel: an unstrained ring is flat, and anything that is not flat is strain or fusion rather than numerical drift.

3.5 Calibrated uncertainty

Conformal intervals hit their guarantee. At a nominal 90%, measured coverage over 5 seeds is 0.871 on ESOL, 0.898 on Lipo and 0.995 on FreeSolv. The FreeSolv figure is over-coverage, not a failure: the guarantee is one-sided.

The interesting question is width. With the default score every molecule receives an interval of the same length, which cannot distinguish a confident prediction from a guess. Conditional coverage shows the cost: on Lipo, coverage drifts by 0.176 across difficulty quintiles — over-covering easy molecules and under-covering hard ones. Scaling the nonconformity score by the Gaussian-process posterior standard deviation cuts that drift to 0.061 and narrows the mean interval from 2.65 to 2.53, at unchanged marginal coverage (0.895). On ESOL the same change makes conditional coverage *worse*, from 0.111 to 0.169. The adaptive score is therefore offered as an option and left off by default.

3.6 Gene Ontology enrichment, and the ordering that breaks it

The panel ships a pruned ontology: biological-process terms with between 5 and 500 annotated genes, 8,098 of 48,340 terms over 18,645 gene symbols, interned as indices to fit the package limit at 2.9 MB. Over-representation is hypergeometric, corrected across terms with Benjamini–Hochberg [4] at FDR = 0.05.

Two conditions matter, and only the second is usually reported. An enrichment must recover biology that is there, and it must stay silent when there is none. The second is where this implementation was wrong. Terms were first dropped for having too few hits, and only the survivors entered the correction — which makes the denominator depend on the outcomes being corrected. Both orderings are run here on identical random sets (Figure 4, left). On 20-gene sets the defective ordering returns a mean of 16.4 significant terms drawn from nothing but noise, as many as 68 in one trial, with at least one term in 35 of 40 trials. Correcting across every term in the window, including the ones that came back empty, gives 0.00.

The defect is invisible at the sizes one is most likely to test with: by 320-gene sets the two orderings agree (0.07 against 0.03). It bites hardest on short, focused lists,

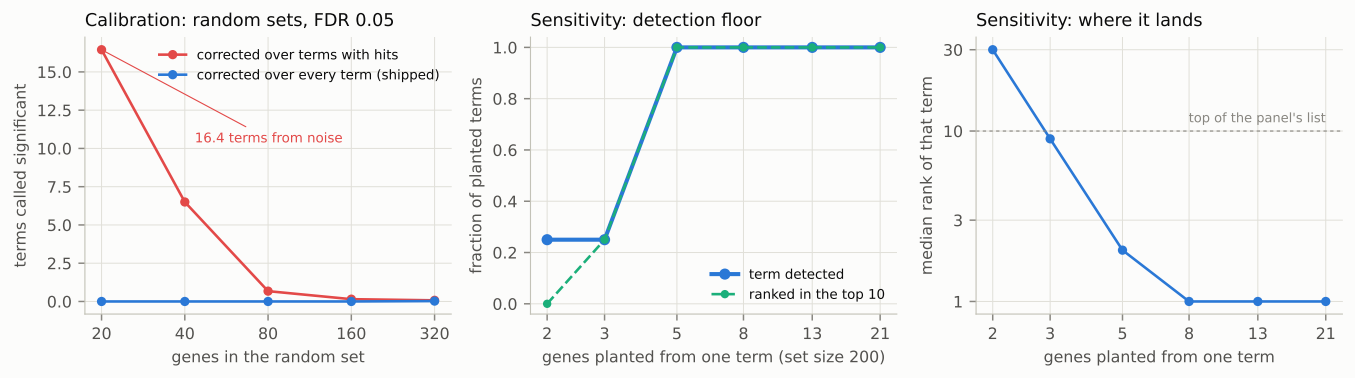


Figure 4: Left: terms called significant on random gene sets, which should be none. Correcting over only the terms that happen to have hits is well calibrated on large sets and badly miscalibrated on small ones, which is where enrichment is usually run. Centre and right: detection and rank for a term planted into a 200-gene set.

because those leave few terms with any hits at all and the correction's denominator collapses onto the handful most likely to look significant. Across every set size, the shipped ordering never exceeds 0.03 terms per random set.

Sensitivity is measured by planting k genes from one term into a 200-gene set, over 12 terms (Figure 4, centre and right). The floor is sharp: 5 planted genes are enough for every term to be detected and ranked inside the top ten, while 2 are recovered only 25.0% of the time and land near rank thirty, far below anything a reader would scroll to. Diluting a whole term, rather than shrinking it, barely registers — terms still rank first with most of the set replaced by noise — so the floor, not a dilution curve, is the honest statement of what the method needs.

Applied to the kinase inhibitor map, 302 of 410 target names resolve, collapsing to 243 distinct genes — a panel writes one kinase under several names, so the two counts are reported separately rather than conflated. Names are resolved by transformations that are accepted only when they land on a real gene, so a mutation suffix (`KIT(D816V)`), a phospho form (`ABL1p`) or a complex (`CDK4-cyclinD1`) reaches its parent gene and nothing else can be silently substituted. The remainder are previous HGNC symbols such as `ASK1` and `ARK5`; they are reported as unmatched rather than guessed at, since a mapping invented to raise the match rate is indistinguishable in the output from biology that was measured.

4 Three failures worth reporting

4.1 A validation set with nothing in it

The scaffold splitter assigns whole scaffold groups to train, validation and test so no scaffold spans two subsets. On FreeSolv it assigned the validation set *zero* molecules. The acyclic group — every molecule with no ring shares the empty scaffold — is 320 of 642 molecules. It overflows the training quota, overflows the validation quota, and lands in test; after that every remaining group still fits in training, so validation is never reached. The split came out 0.50/0.00/0.50 instead of 0.60/0.20/0.20, which silently disabled early stopping and left half the dataset in test.

Nothing raised an error. The benchmark ran, produced plausible RMSEs, and wrote them to disk. The bug surfaced only because an unrelated experiment asked the splitter for a calibration set and got an empty array. Oversized groups are now assigned largest-first, so the group too large for any held-out subset inflates training rather than test, and a rebalancing pass moves whole groups into an undersized validation set. FreeSolv now splits 0.56/0.20/0.24. Results computed before the fix are not comparable with results after it. The MoleculeNet benchmark in Section 3.2 is therefore being regenerated under the corrected splitter; the pre-fix outputs are kept in `results/molnet_presplitfix/` so the two can be compared rather than quietly replaced.

4.2 A generator that lost to its own control

The graph diffusion model scored 2.0% validity — and the untrained marginal baseline, which samples atoms and bonds from the dataset frequencies with no model at all, scored higher. A trained model losing to its own control is not a weak result; it is a broken one, and it was traced rather than written up.

Two bugs, compounding. "No bond" is 83.7% of atom pairs in a molecular graph, so under an unweighted cross entropy, predicting "none" everywhere is close to optimal; the denoiser had learned exactly that, producing graphs of mean degree 0.26 against a training-set mean of 1.99. Separately, the valence-constrained sampler began reverse diffusion from the all-zero bond matrix rather than the marginal noise prior, and since the reverse posterior's likelihood term anchors x_{t-1} to x_t, starting empty dragged the entire trajectory back towards empty.

With a class-balanced bond loss (exponent 1.5, bond weight 3.0) and prior-initialised sampling, trained to 24,000 steps on 6,266 molecules, strict validity reaches 64.7% against 2.0% for the marginal baseline — a model that was

sampler	valid	largest frag.	degree
model + valence	64.7%	100.0%	2.36
model, unconstrained	2.0%	6.0%	4.65
marginal baseline	2.0%	43.0%	1.88
training set	—	—	1.99

Table 1: Generation. Two validity conventions are reported because the literature uses both: a single connected valence-correct molecule, and the largest connected fragment. Mean degree against the training mean is the diagnostic that exposed the collapse.

losing to its own control now beats it by a factor of thirty (Table 1). Of the valid molecules, 100.0% are distinct and 99.5% do not appear in the training set, so the gain is not memorisation. Sampling the same trained model *without* the valence constraint gives 2.0% at mean degree 4.65, nearly all of it rejected for over-valence, so the constraint is doing real work rather than masking a weak denoiser.

The correction overshot slightly: generated mean degree is 2.36 against a training mean of 1.99. The class weighting that lifted the model off the empty graph now pushes it a little past the data, which is worth stating because it is the same knob and the same failure mode in the other direction.

What remains wrong is specific, and it is now the only thing wrong: 35.3% of samples are rejected for being in two or more pieces, and 0.0% for valence errors — every remaining rejection is a connectivity failure. The model has no connectivity prior and the training objective supplies none. That is a single, nameable gap rather than a diffuse weakness, which is the most useful thing a negative result can leave behind. That is the next thing to fix, and it is stated here rather than hidden behind the lenient fragment-validity convention. These hyperparameters were selected on the validity metric reported above, with no held-out set, so that figure should be read as an upper bound.

4.3 An improvement that did not improve

The standard construction for adaptive conformal intervals [14] fits a separate model to the residuals and uses its prediction as the difficulty estimate. Implemented here — an inner fold held out of training, a second Gaussian process regressed on $\log|r|$ — it is worse than the constant-width baseline on every dataset tried: conditional-coverage drift of 0.321 on ESOL, 0.294 on Lipo and 0.219 on FreeSolv, against 0.111, 0.176 and 0.020 for doing nothing at all, with wider intervals throughout. It is reported, and not shipped.

5 Integration with Mantis

The pipeline ships as a packaged extension for Mantis, a map interface for exploring large collections. Installing it modifies nothing: the package goes in through the application's own Import Extension dialog, which validates the manifest, states the permissions and the backend actions, and asks for consent. An earlier version of this work copied three files into the Mantis source and documented three registrations as diffs; the extension mechanism makes that unnecessary, and a reader can install the result without a patch.

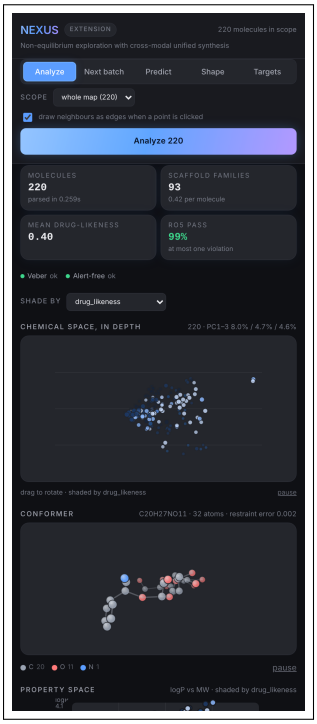


Figure 5: The panel on 220 molecules. The rotating projection carries the map's own fingerprint PCA to a third component; the conformer below it is a distance-geometry embedding relaxed under the out-of-plane term. Both are SVG, so the package stays small enough to ship as text.

The package is a sandboxed panel, a worker host, and a Python backend that is the `nexus` package itself, vendored beside its entry point. There is no service to run and no port to configure. Mantis caps a backend invocation at thirty seconds, so each action bounds its own input and reports what it dropped rather than truncating in silence.

Results return to the map, not to a table. The extension API has no call for colouring points, which turned out to be a useful constraint. Results are instead expressed in the vocabulary the application already has: a scaffold family becomes a named bag and is highlighted, a proposed assay batch becomes the selection, and the Tanimoto neighbours of a clicked molecule become an edge overlay the panel owns and releases when it closes. A chemist reading the map sees the answer in the same terms as everything else on it.

What the integration cost. Two failures are worth recording because neither produces an error. The panel's chart components use utility classes that the host supplies

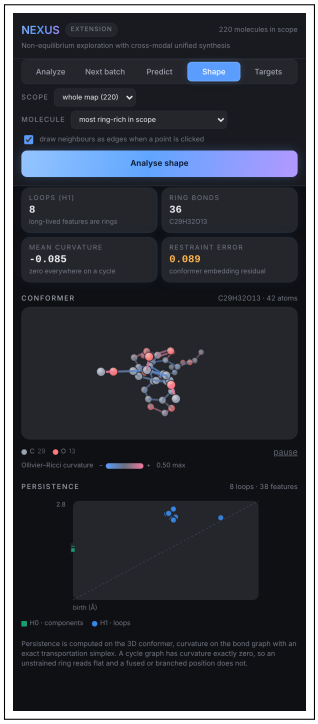


Figure 6: Shape mode. Bonds are coloured by Ollivier–Ricci curvature on a diverging scale centred at zero, beside the persistence diagram of the same conformer: eight loops for a structure with thirty-six ring bonds.

in-tree and an extension iframe does not, so every class was inert and the layout collapsed into run-together text; the package now ships the enumerated set those components reference, and the build fails if one is missing. And inside the dock the panel iframe paints its background while every element in it measures 0×0 — layout is suppressed for the whole document, `html` included. That is not the stylesheet, whose computed height resolves correctly, and not this package: the application's own reference extension does not render there either. Installation, permissions, backend dispatch and SDK access are verified in the running application; the figures above are the same bundle and the same Python driven through a harness.

6 Limitations

- Every implementation here is slower than its established counterpart, often by a large factor. Nothing in this work is a performance claim.
- The aromaticity and valence model is a documented subset, not a complete perception engine. Molecules outside it are reported invalid rather than guessed at.
- The generator produces disconnected graphs in the large majority of samples. Fragment-validity hides this; strict validity does not, which is why both are reported.
- Binding affinity is evaluated on one kinase panel. Cold-drug generalisation is measured, but on a single assay family.
- Generation hyperparameters were selected on the reported metric with no held-out set.
- Conformal guarantees are marginal. Conditional coverage is measured here because the marginal guarantee alone can disguise systematic over-confidence on the hardest cases, but no method here guarantees it.

7 Conclusion

A dependency-free reimplementation is not a contribution in itself, and this paper does not argue otherwise. What the exercise produced was a set of checks tight enough to catch three failures that would otherwise have passed silently: a split that emptied a validation set, a generator beaten by its own control, and an interval that met its guarantee while carrying no information. Each was found by comparing a number against something external — a published table, an untrained baseline, an analytically known answer — rather than by reading code. That comparison is cheap, and it is the part of this work that generalises.

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